Real-Time Intrinsic TND valuation model – project description

IN 22-21 – Times Series Analysis

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1. Project objectives

This project aims to deliver a real-time **intrinsic value model for TND** that blends a fundamental, basket-based exchange rate with adjustments for local market conditions. By doing so, it will provide a continuously updating fair-value estimate for the TND rate, helping to identify when the market is deviating due to liquidity pressures or other frictions.

2. Background and context

The Tunisian Dinar (TND) exchange rate against the US Dollar (USD) is officially determined by the Central Bank of Tunisia (BCT) and is published as an official fixing rate twice daily (once in the morning and once in the evening). These **official TND fixing rates** serve as benchmarks for the market and are typically derived in reference to global currency movements and policy considerations. In particular, the TND's value is influenced by major currency pairs such as EUR/USD, GBP/USD, and USD/JPY. For example, the intrinsic value of USD/TND is treated as one driven by a weighted basket of these major currencies' movements. Conceptually:

$$USD/TND = (Latest\ fixing) \times (\ w_1 \times \%\Delta\ EUR/USD + w_2 \times \%\Delta\ GBP/USD + w_3 \times \%\Delta\ USD/JPY)$$

where w_i is the weight of each currency pair. (*The specific weights or functional form of the equation are not predetermined and will be estimated as part of this project.*)

This formula provides a baseline "intrinsic" value of USD/TND **assuming no local market frictions**, purely reflecting Tunisia's currency value as implied by global currency movements.

3. Problem statement

In practice, the **actual market TND exchange rates** can and does deviate from the above basket-implied value. Local foreign exchange (FX) market conditions — such as liquidity constraints and supply/demand imbalances — cause the market TND rate to diverge from the basket value derived from global currency movements.

Key symptoms of this problem include:

- The TND rate observed in the market may be consistently higher or lower than the fixing. These gaps indicate local **Tunisian FX liquidity pressure** (e.g. scarcity of liquidity tends to raise the TND rate above the theoretical value).

- The magnitude of deviation changes over time and intraday, reflecting real **interbank activity**. For example, during periods of high demand for currency or low liquidity in the Tunisian market, TND might trade at a **premium** relative to the basket-implied rate.

This divergence between the market rate and the basket rate presents a challenge for valuation and risk management. Relying solely on the basket formula could misprice the dinar's true market value. Therefore, the project is motivated by the need to capture and model these deviations in real time, yielding a more accurate "intrinsic value" of TND that accounts for both global fundamentals and local market conditions.

4. Clarification on Interbank (IB) exchange rate data

A critical concept in this project is the **Interbank USD/TND exchange rate**. In our context, the IB rate is the actual FX rate at which banks in Tunisia trade TND with one another. This interbank exchange rate is essentially the true market price for TND under prevailing local liquidity conditions.

Important points about the IB rate data:

- The Central Bank of Tunisia **publishes the average interbank TND exchange rates with a one-day lag**. In other words, each day's reported IB rate reflects the average rate at which banks transacted USD/TND on the previous trading day.
- The IB rate is our **best available proxy for the real market exchange rate** at any given time, but it is not available contemporaneously for intraday modeling. The delay means we only see today what yesterday's interbank rate was, leaving a gap in real-time insight.
- The IB rate captures the effects of local market pressures.

Understanding these points, you must incorporate interbank rate information carefully. The difference between the official BCT fixing and the delayed IB rate serves as a measurable indicator of **market frictions and liquidity pressures**. However, since our goal is a real-time model, we need to develop methods to **infer or forecast the IB rate during the current day** (i.e. to *nowcast* the IB rate) in order to adjust the basket value in real time. This also implies modeling the adjustment between the interbank rate and the official fixing rate. Since the BCT publishes two fixings per day (morning and evening), but the IB is published one day later, one research question is which fixing to use when modeling the adjustment — today's morning, evening, or yesterday's. The appropriate approach is left for you to determine.

5. Modeling approach: Baseline + stochastic liquidity adjustment

The core idea for the intrinsic valuation model is to combine two components:

1. **Baseline Value (Basket-Based):** This is the TND value derived from the official fixing and global currency movements. It serves as an anchor reflecting the fundamental value if the Tunisian dinar were to move in line with major currencies. At the start of a period (morning or evening), the model uses the latest BCT fixing as the anchor point. As global markets move (fluctuations in EUR/USD, GBP/USD, USD/JPY), the baseline is updated continuously via the function $f(\cdot)$ to reflect what TND should be absent local pressures. This component ensures the model reacts to real-time international market information.

2. **Stochastic adjustment for local market conditions:** This second component is a dynamic adjustment factor capturing the deviation caused by local FX liquidity and market frictions. It

represents how far the *actual* TND rate is likely to stray from the baseline at a given moment. The adjustment can be modeled as a stochastic process – for instance, a mean-reverting spread or a random walk – to reflect real time supply/demand shocks, liquidity shortages, or surpluses. Key considerations for modeling this adjustment include:

- **Historical spread analysis:** Analyze historical data to observe how the spread between the interbank rate and the fixings behaves. This spread quantifies the typical local deviation over time.
- **Statistical modeling:** Calibrate a statistical or machine-learning model that uses the historical spread behavior to nowcast the current deviation in real time. For example, an error-correction model or Kalman filter could update the estimated spread as new partial information comes in during the day.
- **Dealing with data lag:** Because the IB rate is reported with delay, incorporate nowcasting techniques to infer today's deviation. You may use the previous day's observed spread as a starting point each morning, then adjust it using any intraday signals available (e.g. live indicative quotes from major banks or patterns observed around the time of fixings).
- **Intraday regime shifts:** Account for the possibility that liquidity conditions vary over the course of the day. The model might allow the volatility or bias of the adjustment factor to change between, say, morning and afternoon trading sessions (for example, perhaps deviations tend to widen toward end-of-day when liquidity thins).

In summary, the model will output a **real-time intrinsic TND rates** computed as:

This intrinsic value should closely track where TND would trade if one accounts for both international currency fundamentals and current local market pressure. It provides a continuously updating "fair value" estimate for the exchange rate.

6. Research tasks and deliverables

The project will proceed with several research and development tasks. The key deliverables expected from the team include a **word document** and an **Excel sheet** (and/or Python code) that perform the following:

- **Model framework design:** A comprehensive design document for the model structure, clearly detailing the two components (basket-based baseline and liquidity adjustment) and how they combine. This should include the mathematical formulation and a description of how the model will operate in real time.
- **Basket weight estimation:** Using historical data, estimate and validate the function f that links changes in EUR/USD, GBP/USD, and USD/JPY to movements in USD/TND. This entails determining the appropriate weights or coefficients that best replicate the BCT's fixing behavior. The deliverable will include the estimated weights/parameters and an explanation of the estimation process (e.g. regression methodology and goodness-of-fit).
- **Interbank Rate Nowcasting Mechanism:** Develop an approach to nowcast the interbank USD/TND exchange rate (or equivalently, the liquidity-driven deviation) during the trading day. This could involve statistical time-series techniques, machine learning models, or heuristic rules, and should leverage any available intraday information. The deliverable will be a prototype or algorithm that produces an estimate of the current USD/TND market rate (or the deviation from the baseline) in real time, despite the official data lag.
- **Stochastic adjustment integration:** Integrate the nowcasted liquidity adjustment into the overall model. This includes implementing the logic that adjusts the baseline rate by the estimated deviation at each moment. The deliverable should explain how this integration

works in practice (e.g. update frequency, handling of volatility or sudden jumps in the spread) and ensure that the combined output remains stable and realistic.

- **Real-Time Intrinsic value output:** Build a prototype system or tool that generates the intrinsic USD/TND value stream in real or near-real time. This might be a script or application that continuously ingests live data (the relevant global FX rates and any local market indicators) and outputs the calculated intrinsic rate. The team should demonstrate this system on recent data, showing that it can operate continuously and handle edge cases or data interruptions gracefully.
- **Performance evaluation and backtesting:** Provide a thorough evaluation of the model's performance. Using historical periods (including out-of-sample tests), backtest the intrinsic value estimates against actual market outcomes (such as the next day's published interbank rate or any known intraday market rate data). This deliverable should include error statistics, correlation analysis, and visual charts to illustrate how closely the model's output tracks reality. Additionally, discuss scenarios where the model struggled and any adjustments made to improve it. This validation step will build confidence in the model's reliability and highlight any limitations.

Each deliverable should be accompanied by clear documentation.

7. Appendix

This appendix defines all financial and market concepts referenced implicitly or explicitly in the project description. Definitions follow standard usage in central banking, foreign exchange markets, and applied financial econometrics.

- **Foreign Exchange (FX) Market:** The market in which currencies are exchanged. Participants include central banks, commercial banks, corporations, and financial institutions.
- **Currency Pair (X/Y):** The number of units of currency **Y** required to purchase one unit of currency **X** (Exp: USD/TND = 2.85 means 1 USD costs 2.85 TND).
- **Intrinsic Value (Exchange Rate):** An estimated exchange rate that reflects the underlying or fundamental value of a currency, abstracting from temporary distortions and market frictions.
- **Market Frictions:** Structural or temporary factors (like liquidity constraints, transaction costs, or regulatory limits) that prevent the exchange rate from perfectly reflecting fundamentals.
- **Liquidity:** The degree to which foreign currency can be bought or sold in the market without causing large or abrupt changes in the exchange rate. It reflects the availability of foreign currency and the balance between supply and demand among market participants.
- **Premium:** A situation in which the market exchange rate trades above a reference or benchmark rate.
- **Discount:** A situation in which the market exchange rate trades below a reference or benchmark rate.
- **Kalman Filter:** A recursive state-space estimation method used to estimate unobserved (latent) variables over time by combining a dynamic system model with noisy or delayed observations.

State equation (transition):

$$x_t = A\, x_{\{t-1\}} + w_t \qquad \mathrm{w_t} \sim \mathrm{N}(0, \mathrm{Q})$$

Observation equation (measurement):

$$y_t = H\, x_t + v_t \qquad\qquad v_t \sim N(0, R)$$

where:

- x_t is the unobserved state variable
- y_t is the observed variable
- A is the state transition matrix
- H is the observation matrix
- w_t, v_t are mutually independent noise terms